Essentials of
economics

Krugman • Wells • Olney

chapter 4

The Market Strikes Back

d by:

Yvonn Quijano

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BIG CITY, NOT-SO-BRIGHT IDEAS

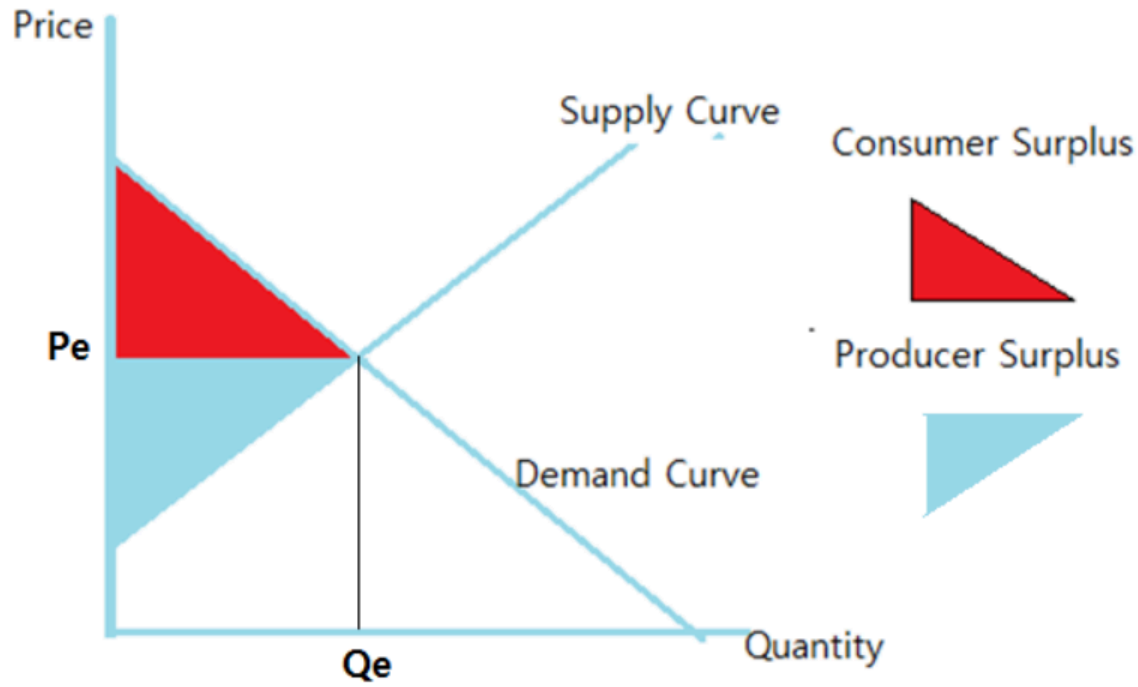


New York City: An empty taxi is hard to find.

What you will learn in this chapter:

- The meaning of **price controls** and **quantity controls**, two kinds of government intervention in markets
- How price and quantity controls create problems and make a market **inefficient**
- Why economists are often deeply skeptical of attempts to intervene in markets
- Who benefits and who loses from market interventions, and why they are used despite their well- known problems
- What an **excise tax** is and why its effect is similar to a quantity control
- Why the **deadweight loss** of a tax means that its true cost is more than the amount of tax revenue collected

Consumer Surplus and Producer Surplus



Welfare: The benefits that a society can get

Welfare of the competitive market = consumer surplus + producer surplus

Why surplus? See the textbook p.106, and p.112

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Consumer Surplus: "Willingness to Pay"

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TABLE 4-1 Consumer Surplus When the Price of a Used Textbook Is $30

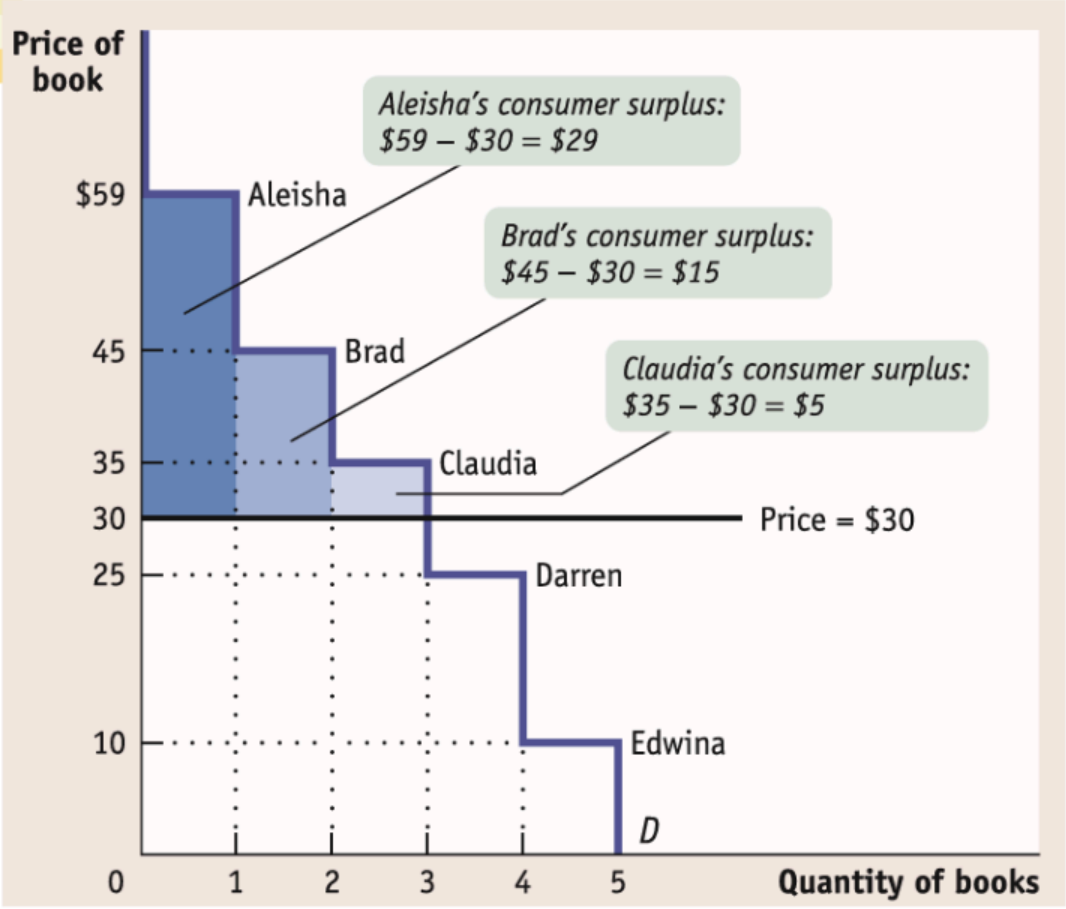
Potential buyer	Willingness to pay	Price paid	Individual consumer surplus = Willingness to pay − Price paid
Aleisha	$59	$30	$29
Brad	45	30	15
Claudia	35	30	5
Darren	25	—	—
Edwina	10	—	—
All buyers			**Total consumer surplus = $49**

Consumer Surplus: "Willingness to Pay"

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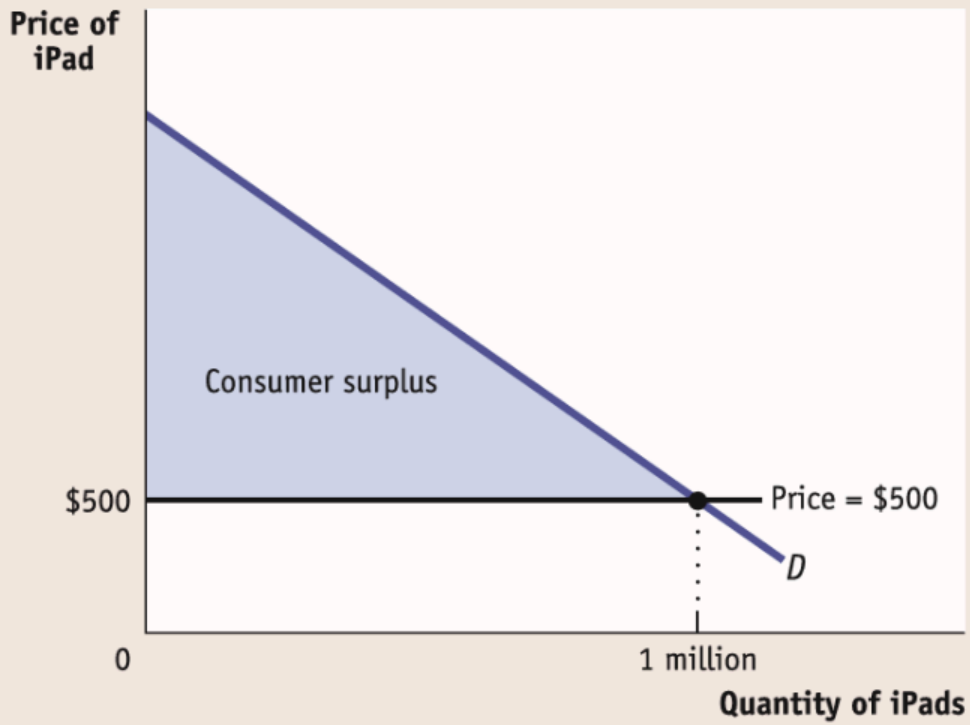


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Consumer Surplus: "Willingness to Pay"

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FIGURE 4-2 Consumer Surplus



The demand curve for iPads is smooth because there are many potential buyers. At a price of $500, 1 million iPads are demanded. The consumer surplus at this price is equal to the shaded area: the area below the demand curve but above the price. This is the total net gain to consumers generated from buying and consuming iPads when the price is $500.

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Producer Surplus: "Willingness to Sell-Cost"

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TABLE 4-2 Producer Surplus When the Price of a Used Textbook Is $30

Potential seller	Cost	Price received	Individual producer surplus = Price received − Cost
Andrew	$5	$30	$25
Betty	15	30	15
Carlos	25	30	5
Donna	35	—	—
Engelbert	45	—	—
All sellers			**Total producer surplus = $45**

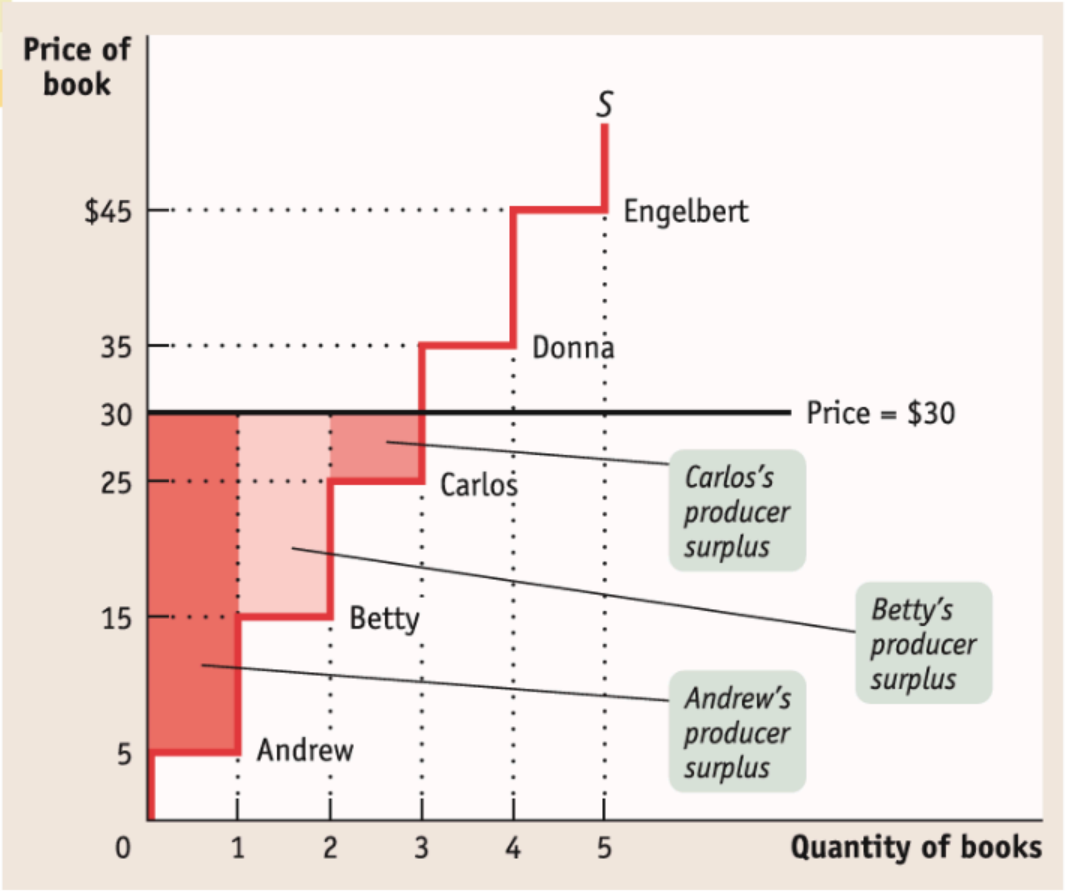
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Engelbert	45	—	—
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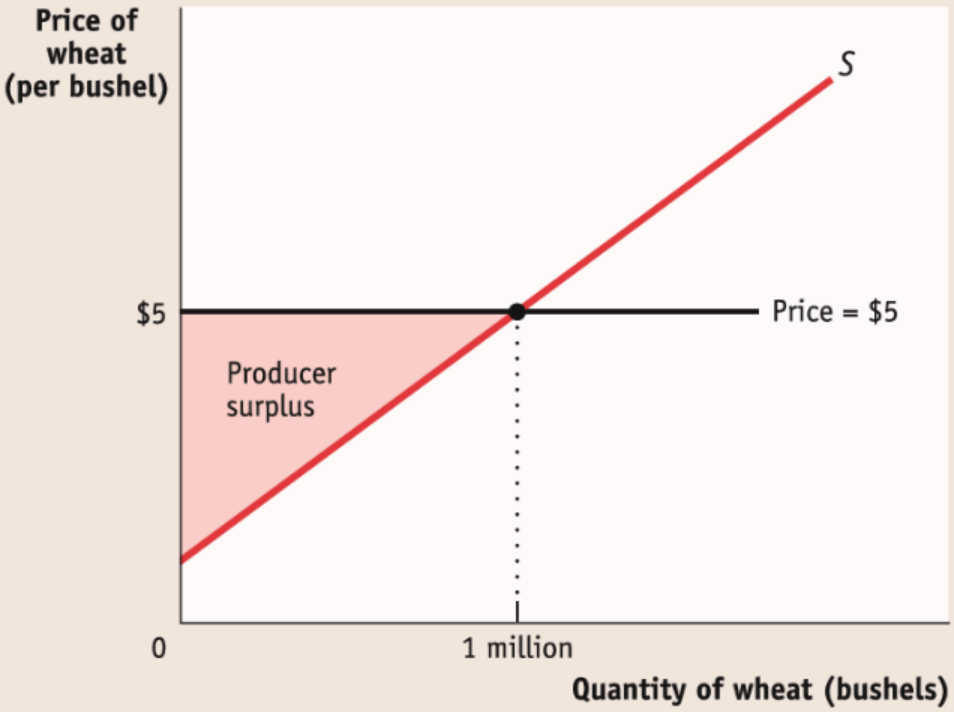
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Producer Surplus

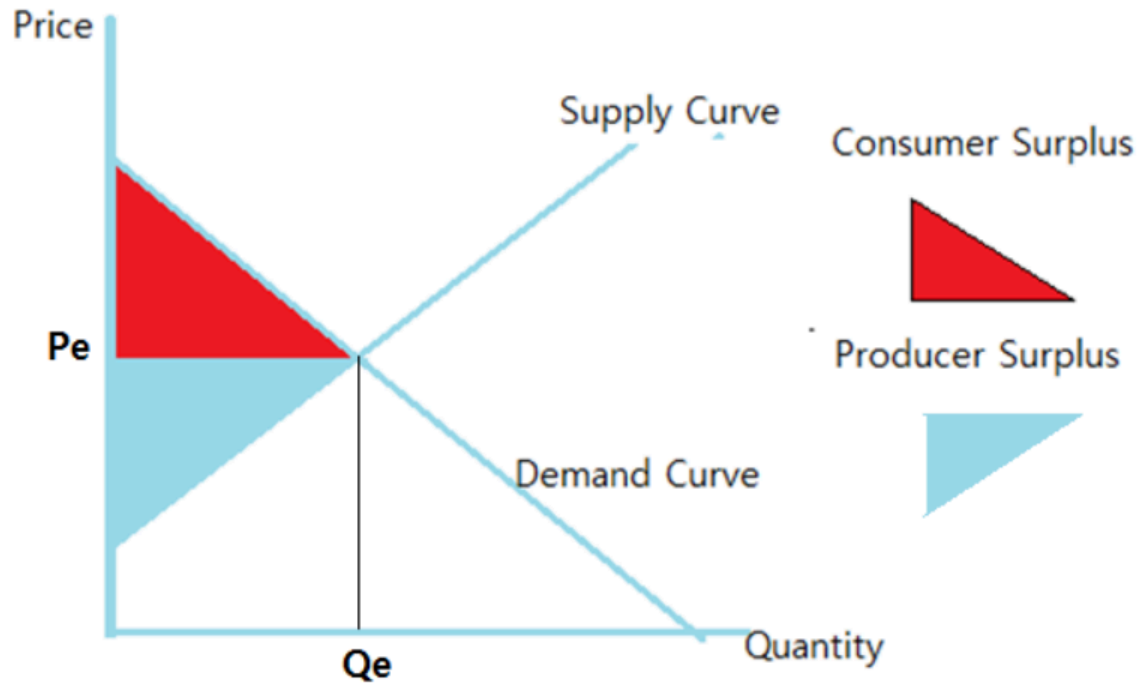
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FIGURE 4-4 Producer Surplus

Here is the supply curve for wheat. At a price of $5 per bushel, farmers supply 1 million bushels. The producer surplus at this price is equal to the shaded area: the area above the supply curve but below the price. This is the total gain to producers—farmers in this case—from supplying their product when the price is $5.



Consumer Surplus and Producer Surplus



Welfare: The benefits that a society can get

Welfare of the competitive market = consumer surplus + producer surplus

Why surplus? See the textbook p.106, and p.112

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Government's Control of Prices or Quantity

Price Control

(1) Price Ceiling: A maximum price that sellers are allowed to charge (lower than the market equilibrium price)

ex) Rent control: a landlord cannot charge to a tenant above a certain level.

(2) Price Floor: A minimum price buyers are required to pay (higher than the market equilibrium price)

ex) Minimum wage: Firms (buyers) must give at least the minimum wage to employees (sellers).

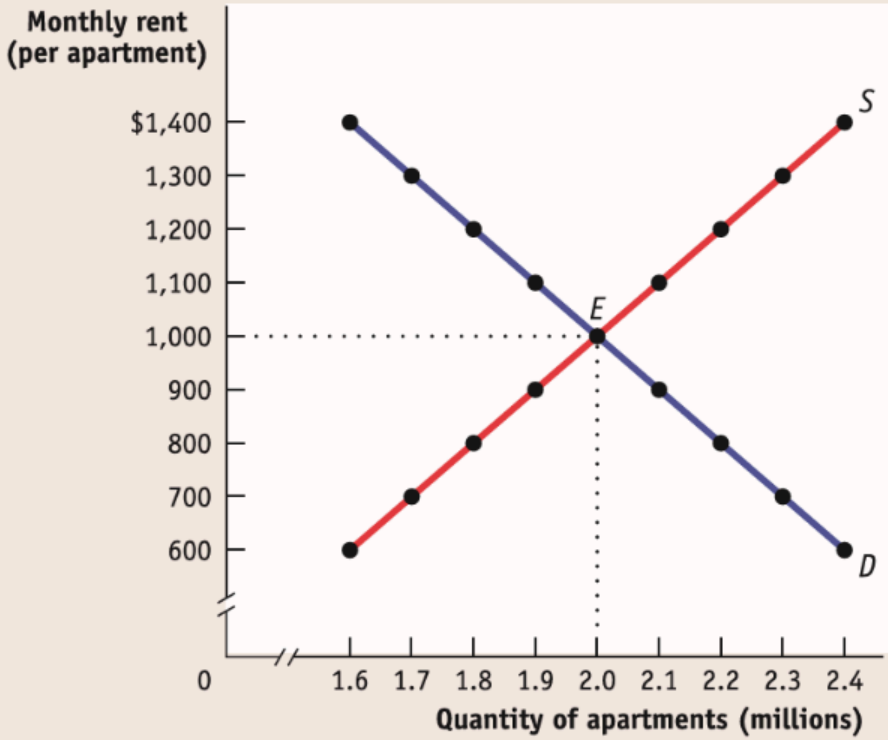
Quantity Control

ex) Taxi licensing

Price Ceilings

Why a Price Ceiling Causes Inefficiency

FIGURE 4-6 The Market for Apartments in the Absence of Price Controls



Monthly rent (per apartment)	Quantity of apartments (millions)	
	Quantity demanded	Quantity supplied
$1,400	1.6	2.4
1,300	1.7	2.3
1,200	1.8	2.2
1,100	1.9	2.1
1,000	2.0	2.0
900	2.1	1.9
800	2.2	1.8
700	2.3	1.7
600	2.4	1.6

Without government intervention, the market for apartments reaches equilibrium at point *E* with a market rent of $1,000 per month and 2 million apartments rented.

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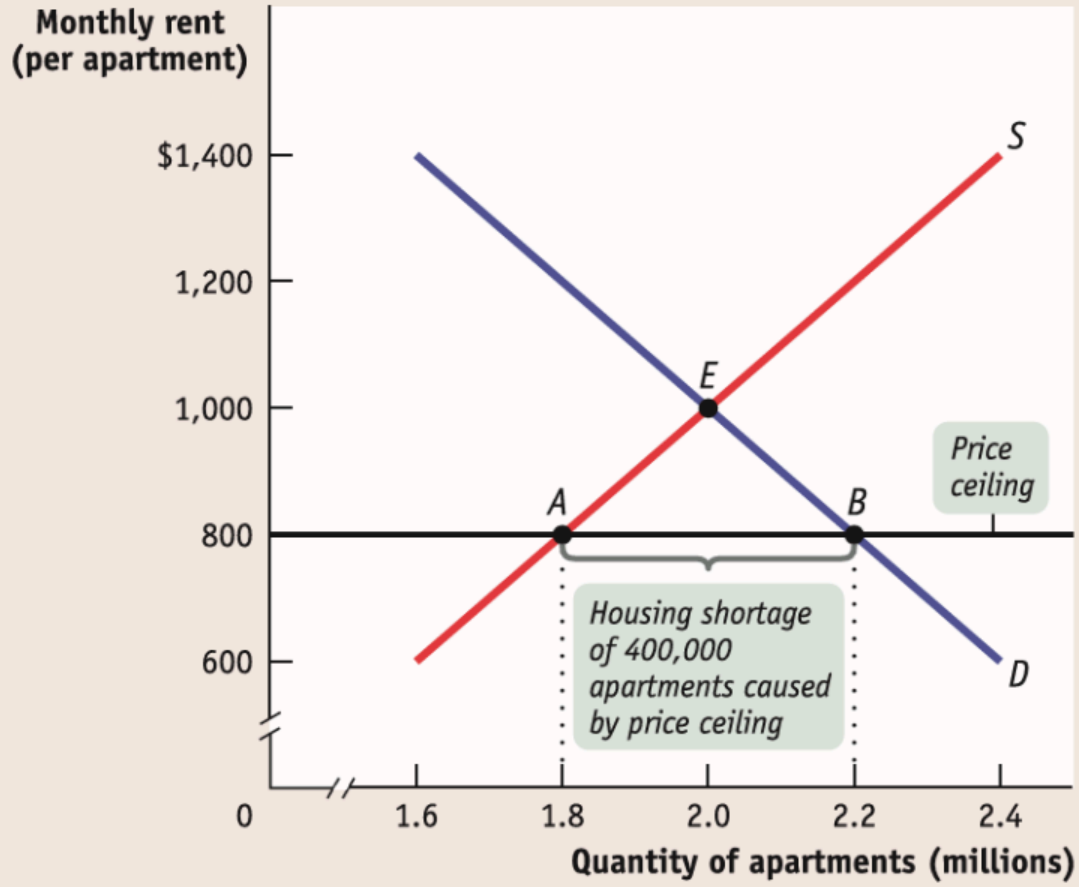
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Price Ceilings

Why a Price Ceiling Causes Inefficiency

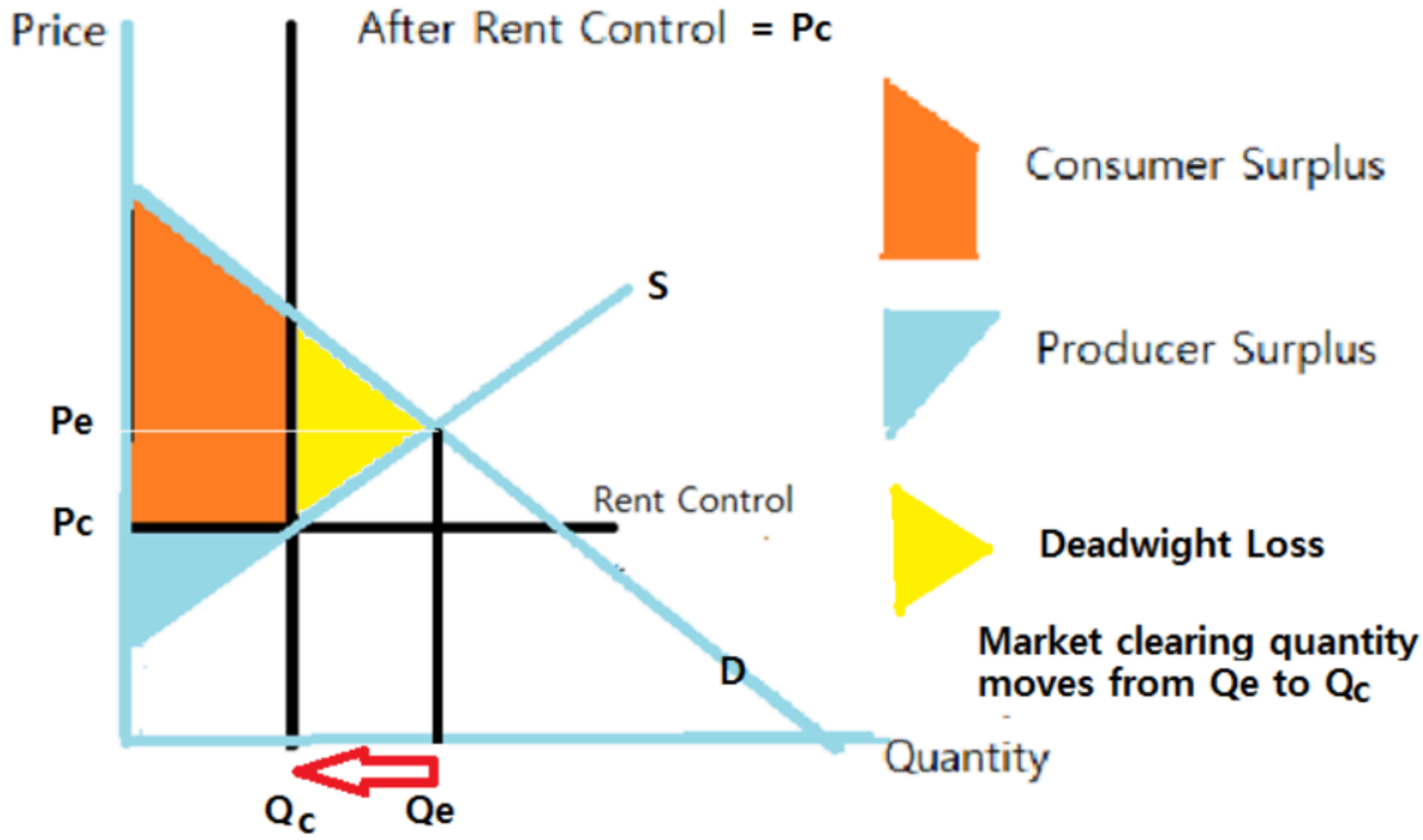
FIGURE 4-7 The Effects of a Price Ceiling

The black horizontal line represents the government-imposed price ceiling on rents of $800 per month. This price ceiling reduces the quantity of apartments supplied to 1.8 million, point *A*, and increases the quantity demanded to 2.2 million, point *B*. This creates a persistent shortage of 400,000 units: 400,000 people who want apartments at the legal rent of $800 but cannot get them.



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Welfare Change after Rent Control



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Price Ceilings

Why a Price Ceiling Causes Inefficiency

A market or an economy is **inefficient** if there are missed opportunities: some people could be made better off without making other people worse off by, for example, subletting.

(1) Inefficient Allocation to Consumers

Price ceilings often lead to inefficiency in the form of **inefficient allocation to consumers**: people who want the good badly and are willing to pay a high price don't get it, and those who care relatively little about the good and are only willing to pay a low price do get it.

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Price Ceilings

(2) Wasted Resources

Price ceilings typically lead to inefficiency in the form of **wasted resources**: people spend money and expend effort in order to deal with the shortages caused by the price ceiling. For the rent control, people will spend much time and money to rent houses.

(3) Inefficiently Low Quality

Price ceilings often lead to inefficiency in that the goods being offered are of **inefficiently low quality**: sellers offer low-quality goods at a low price even though buyers would prefer a higher quality at a higher price.

Price Ceilings

(4) Black Markets can be created

A **black market** is a market in which goods or services are bought and sold illegally —either because it is illegal to sell them at all or because the prices charged are legally prohibited by a price ceiling.

In the case of rent control, landlords can ask extra money to house seekers illegally in addition to the rent

Price Ceilings

So Why Are There Price Ceilings?

We have seen three common results of price ceilings:

- A persistent shortage of the good
- Inefficiency arising from this persistent shortage in the form of inefficient allocation of the good to consumers, resources wasted in searching for the good, and the inefficiently low quality of the good offered for sale
- The emergence of illegal, black market activity

There are people who get benefits from the price ceiling. Sometime, they are well organized and politically strong.

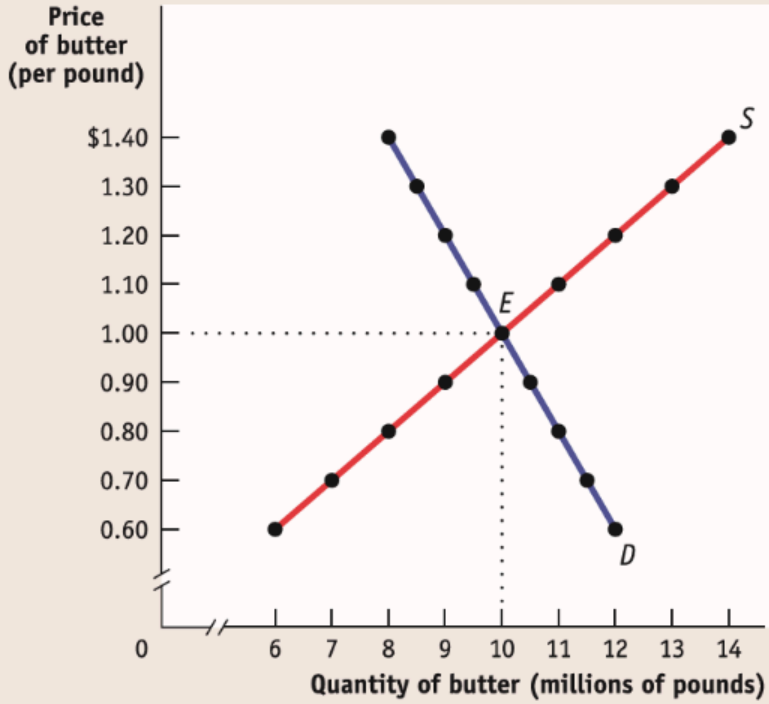
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Price Floors

The **minimum wage** is a legal floor on the wage rate, which is the market price of labor.

Example of a market for butter

FIGURE 4-10 The Market for Butter in the Absence of Government Controls



Price of butter (per pound)	Quantity of butter (millions of pounds)	
	Quantity demanded	Quantity supplied
$1.40	8.0	14.0
1.30	8.5	13.0
1.20	9.0	12.0
1.10	9.5	11.0
1.00	10.0	10.0
0.90	10.5	9.0
0.80	11.0	8.0
0.70	11.5	7.0
0.60	12.0	6.0

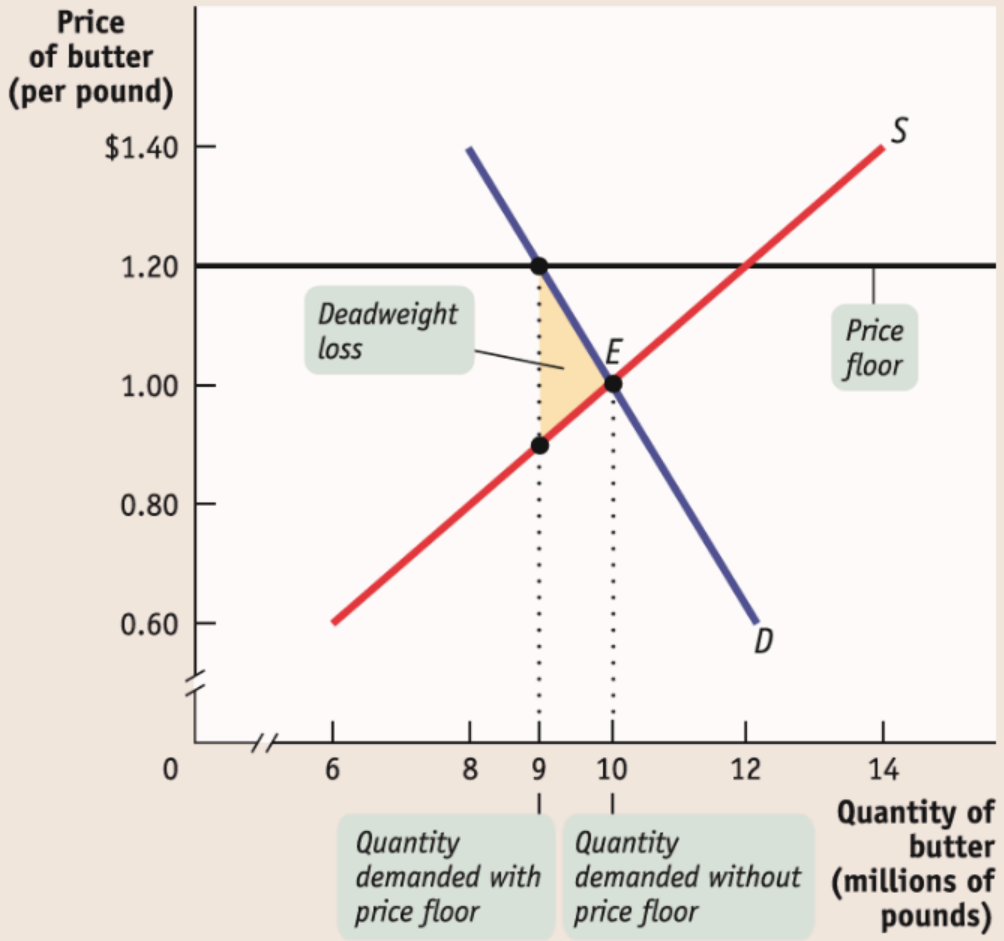
Without government intervention, the market for butter reaches equilibrium at a price of $1 per pound with 10 million pounds of butter bought and sold.

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Price Floors

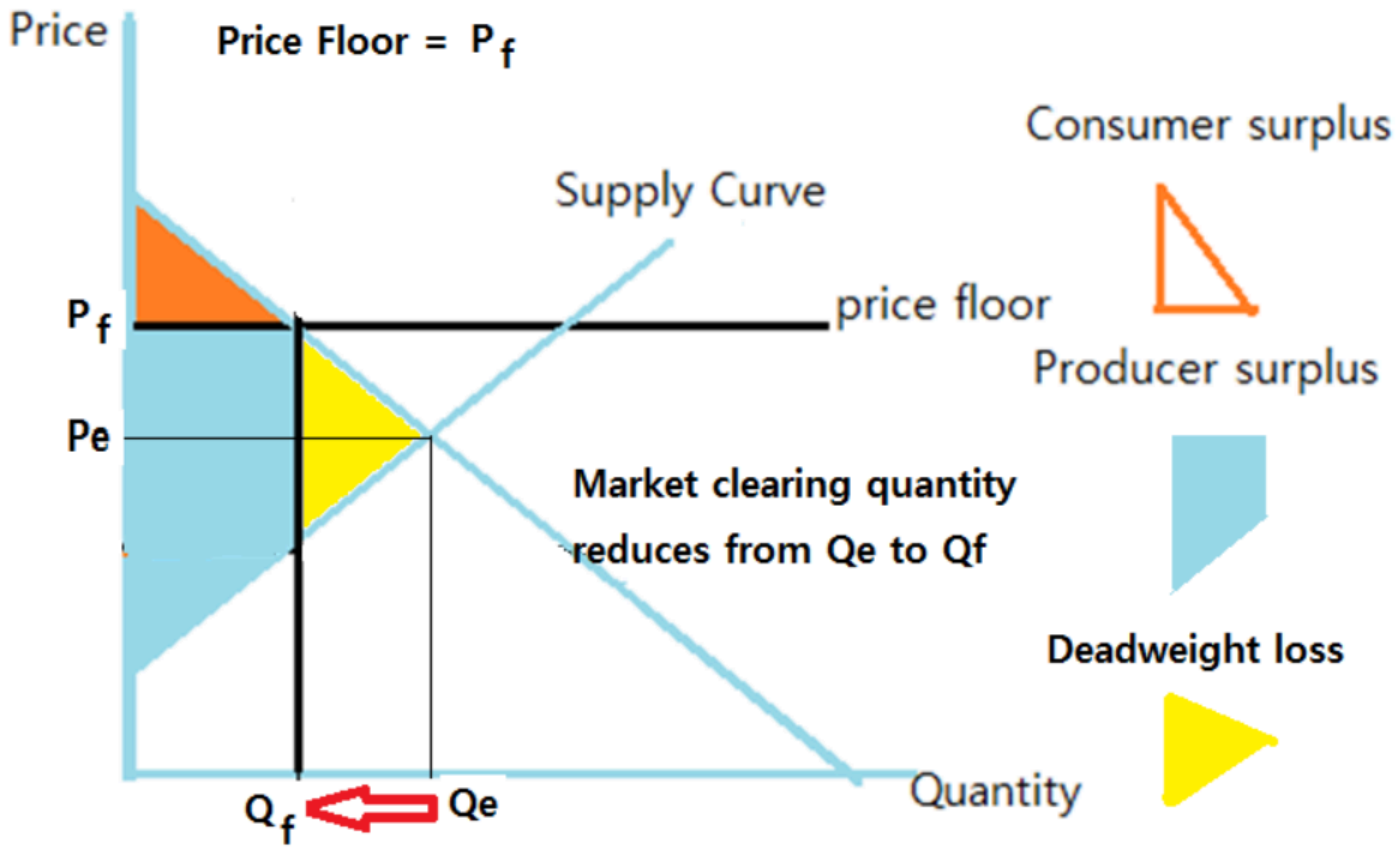
FIGURE 4-12 A Price Floor Causes Inefficiently Low Quantity

A price floor reduces the quantity demanded below the market equilibrium quantity and leads to a deadweight loss.



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Price Floor


Price
Price Floor = P_f
Consumer surplus
Supply Curve
price floor
P_f
Producer surplus
Pe
Market clearing quantity
reduces from Qe to Qf
Deadweight loss
Quantity
Q_f
Qe

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Price Floors

Why a Price Floor Causes Inefficiency

Inefficient Allocation of Sales Among Sellers

Price floors lead to **inefficient allocation of sales among sellers**: those who would be willing to sell the good at the lowest price are not always those who actually manage to sell it.

Wasted Resources

The most graphic examples involve government purchases unwanted surpluses caused by price floors.

Surplus production is sometimes destroyed, which is pure waste

In other cases, the stored produce goes, as officials euphemistically put it, "out of condition" and must be thrown away

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Price Floors

Why a Price Floor Causes Inefficiency

Inefficiently High Quality

Price floors often lead to inefficiency in that goods of **inefficiently high quality** are offered: sellers offer high-quality goods at a high price, even though buyers would prefer a lower quality at a lower price. It tends to be buyer's market. Buyers can choose the sellers they like.

Illegal Activity

Like price ceilings, price floors can provide an incentive for *illegal activity*.

Price Floors

So Why Are There Price Floors?

To sum up, a price floor creates various negative side effects:

- A persistent surplus of the good
- Inefficiency arising from the persistent surplus (from low quantity demanded) in the form of inefficient allocation of sales among sellers, wasted resources, and an inefficiently high level of quality offered by suppliers
- The temptation to engage in illegal activity, particularly bribery and corruption of government officials.

  In the case of minimum wage, firms may hire at a wage lower than the minimum wage and give money to government inspectors.

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Controlling Quantities

A **quantity control**, or **quota**, is an upper limit on the quantity of some good that can be bought or sold. The total amount of the good that can be legally transacted is the **quota limit**.

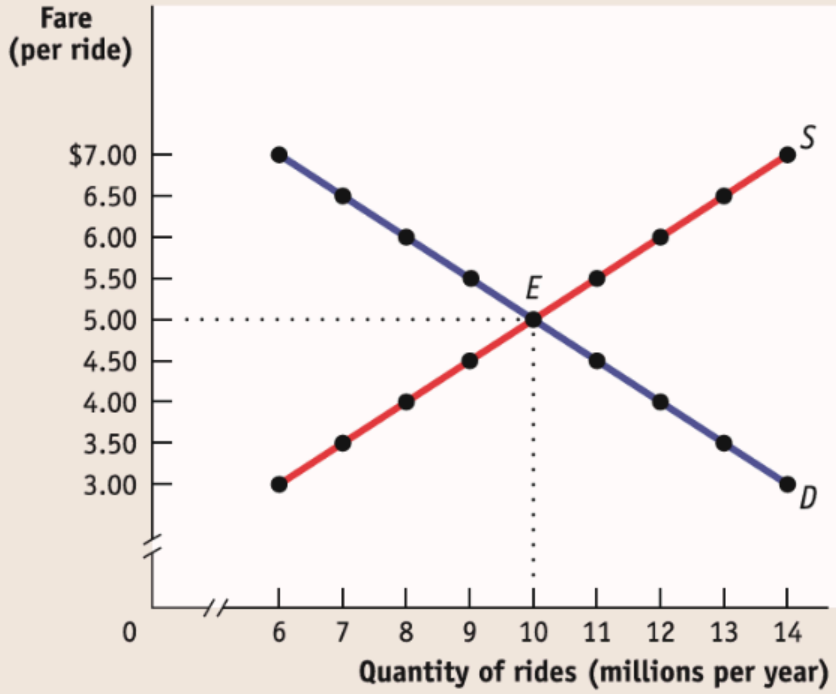
A **license** gives its owner the right to supply a good.

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Controlling Quantities

The Anatomy of Quantity Controls

FIGURE 4-13 The Market for Taxi Rides in the Absence of Government Controls



Fare (per ride)	Quantity of rides (millions per year)	
	Quantity demanded	Quantity supplied
$7.00	6	14
6.50	7	13
6.00	8	12
5.50	9	11
5.00	10	10
4.50	11	9
4.00	12	8
3.50	13	7
3.00	14	6

Without government intervention, the market reaches equilibrium with 10 million rides taken per year at a fare of $5 per ride.

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Controlling Quantities

The Anatomy of Quantity Controls

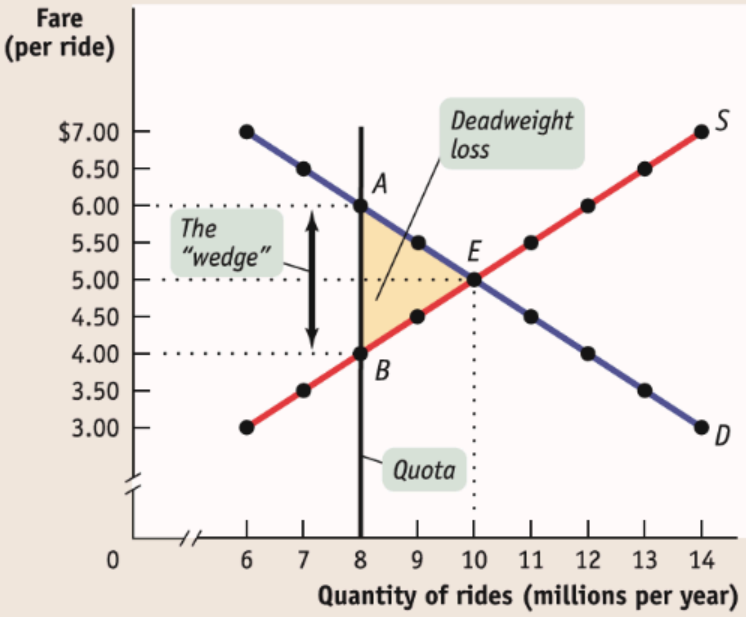
The **demand price** of a given quantity is the price at which consumers will demand that quantity.

The **supply price** of a given quantity is the price at which producers will supply that quantity.

Controlling Quantities

The Anatomy of Quantity Controls

FIGURE 4-14 Effect of a Quota on the Market for Taxi Rides



Fare (per ride)	Quantity of rides (millions per year)	
	Quantity demanded	Quantity supplied
$7.00	6	14
6.50	7	13
6.00	8	12
5.50	9	11
5.00	10	10
4.50	11	9
4.00	12	8
3.50	13	7
3.00	14	6

The table shows the demand price and the supply price corresponding to each quantity: the price at which that quantity would be demanded and supplied, respectively. The city government imposes a quota of 8 million rides by selling licenses for only 8 million rides, represented by the black vertical line. The price paid by consumers rises to $6 per ride, the demand price of 8 million rides, shown by point *A*. The supply price of 8 million rides is only $4 per ride, shown by point *B*. The difference between these two prices is the quota rent per ride, the earnings that accrue to the owner of a license. The quota rent drives a wedge between the demand price and the supply price. And since the quota discourages mutually beneficial transactions, it creates a deadweight loss equal to the shaded triangle.

Controlling Quantities

The Anatomy of Quantity Controls: "Quota Rent"

A quantity control, or quota, drives a **wedge** between the demand price and the supply price of a good; that is, the price paid by buyers ends up being higher than that received by sellers. The difference between the demand and supply price at the quota limit is the **quota rent**, the earnings that accrue to the license-holder from ownership of the right to sell the good. It is equal to the market price of the license when the licenses are traded.

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Quota Rent

(I myself, Mr. A, Mr. B, Mr. C, ……..) would like to earn money by driving a taxi.

We will be all happy if we can get $4.00 from customers for a ride (We will not be happy if we get less than $4.00 a ride). But customers would like to pay us $6.00 because there are not enough taxis.

Right now I am the only person having the right to run a taxi. But today I cannot drive it. So, I want to let other person drive it – I want to rent out to other driver. How much can I get for the rent?

$1.00 from Mr. A? – possible, but Mr. B is going to give me $1.50.
$2.50 – impossible, no one will borrow my taxi because rider pays only $6.00. If he pays me $2.50, he can only earn $3.50 which is less than $4.00.
The rent will be exactly $2.00. I, and the renter both will be happy.

Controlling Quantities

The Costs of Quantity Controls

Quantity controls typically create the following undesirable side effects:

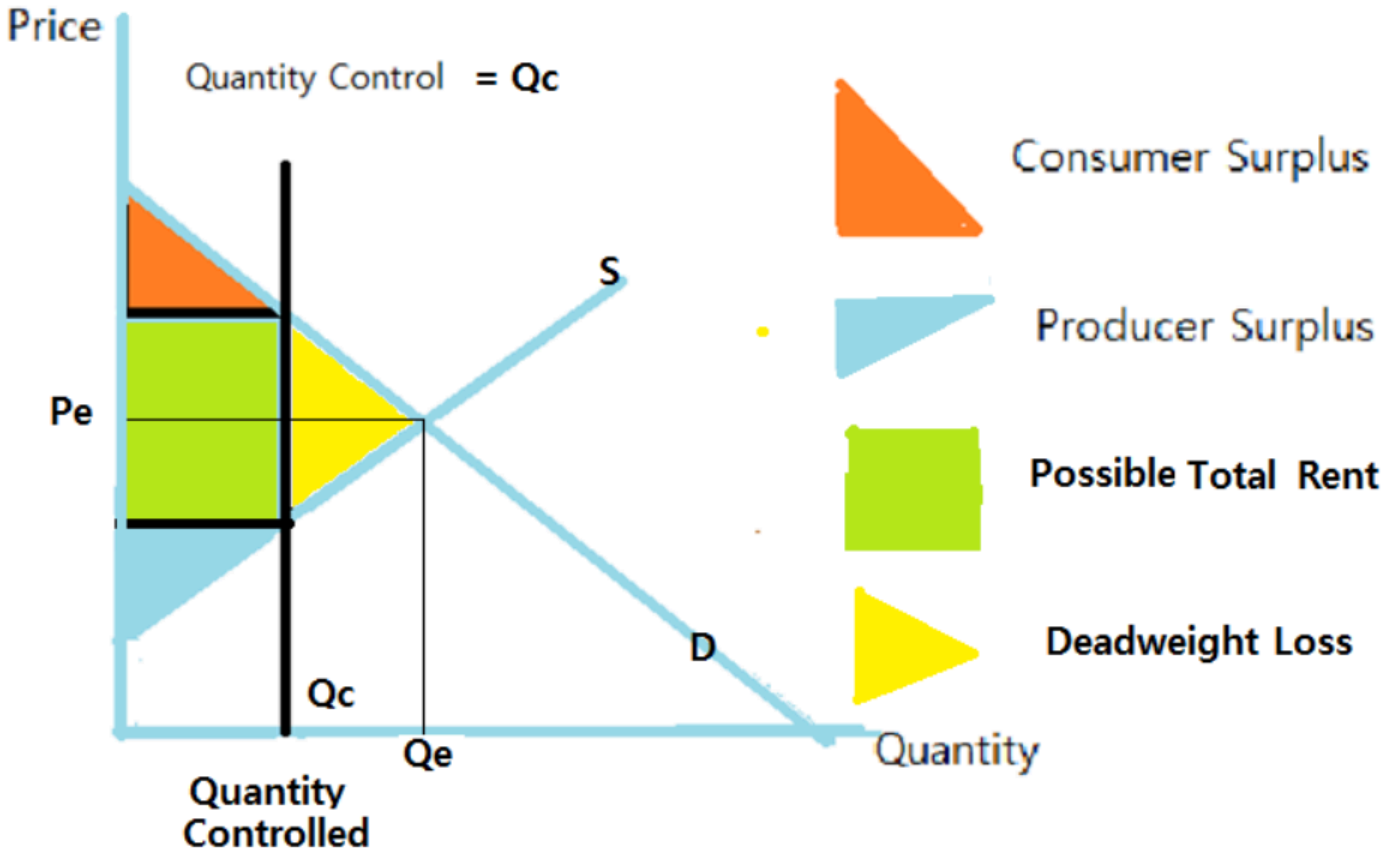
- Inefficiencies, or missed opportunities, in the form of mutually beneficial transactions that don't occur

  There are drivers to provide rides and there are riders willing to pay for it.

- Incentives for illegal activities

  U**nlicensed cars pick up riders which is illegal**.

Quantity Control



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A Surprise Parallel: Taxes

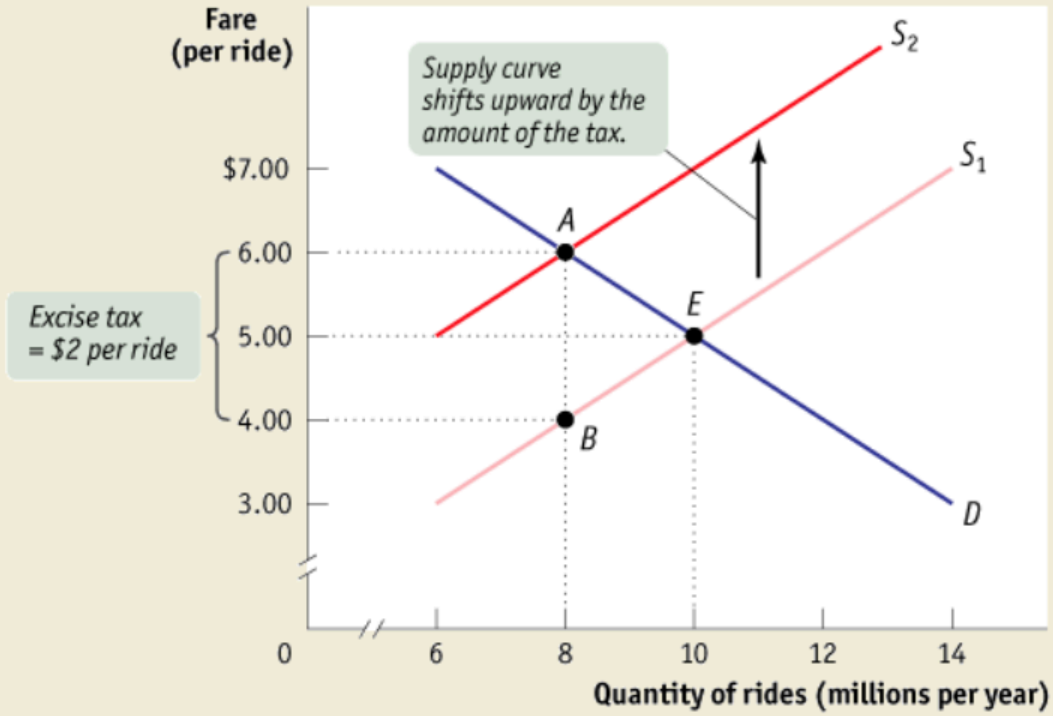
Why Is a Tax like a Quota?

An **excise tax** is a tax on sales of a good or service.

Figure 4-7

Effect of an Excise Tax Levied on Sales of Taxi Rides

S_1 is the supply curve before the tax. After the city requires drivers to pay a tax of $2 for every ride they give, the supply curve shifts upward by $2, to the new supply curve S_2. This means that the price drivers receive net of tax is $4, represented by point *B* on the old supply curve S_1. And the price paid by riders is $6, represented by point *A* on the new supply curve S_2. The tax drives a wedge between the demand price, $6, and the original supply price, $4.



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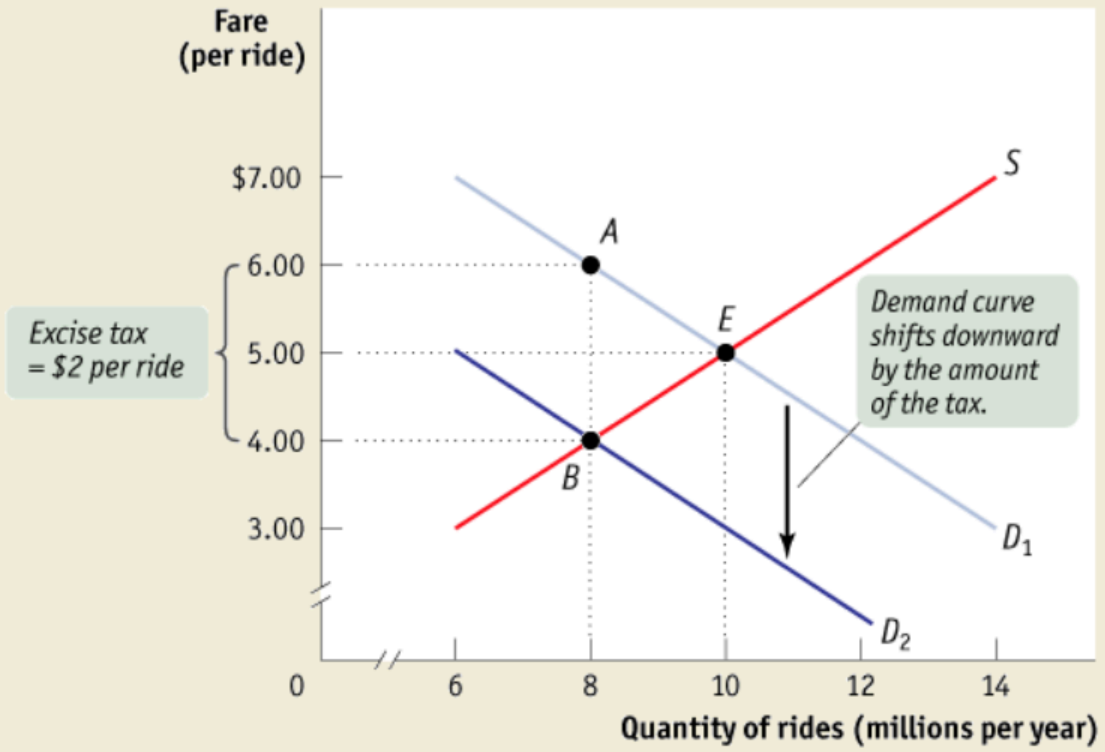
A Surprise Parallel: Taxes

Who Pays an Excise Tax?

Figure 4-8

Effect of an Excise Tax Levied on Purchases of Taxi Rides

D_1 is the demand curve before the tax. After the city requires riders to pay the $2 tax per ride, the demand curve shifts down by $2 to the new demand curve D_2. Drivers again receive, net of tax, $4, represented by point *B*, while riders again pay a total price of $6, represented by point *A*. The incidence of the tax is exactly the same as in Figure 4-7. This shows that who officially pays a tax is irrelevant when answering the question of who bears the burden of the tax.



The **incidence** of a tax is a measure of who really pays it.

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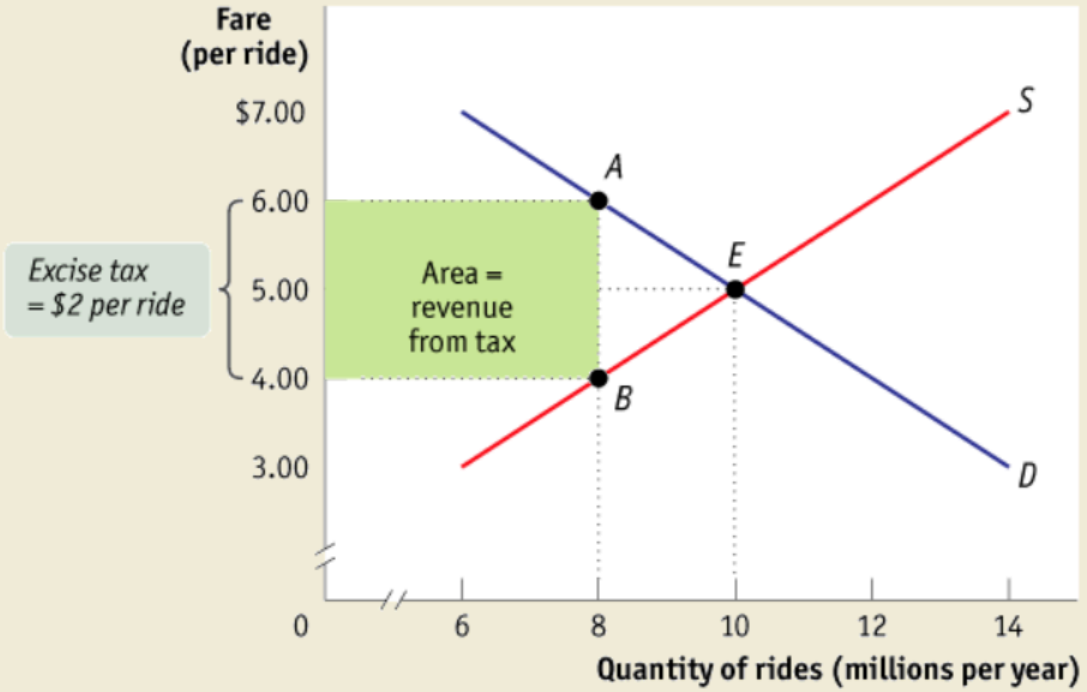
A Surprise Parallel: Taxes

The Revenue from an Excise Tax

Figure 4-9

The Revenue from an Excise Tax

The government revenue collected by this excise tax is equal to the area of the shaded rectangle. In this case it is $2 per ride × 8 million rides = $16 million.



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Producer pays, Consumer pays

$4.00 + $2.00 tax

Producer pays the tax:
- Producer collects $6.00
- Government collects $2.00
- Producer real earnings $4.00
- Consumer pays $6.00

Consumer pays the tax:
- Consumer pays $6,00
- Producer earnings $4.00
- Government collects $2.00

The same!

chapter 4

Tax Incidence

What is tax incidence?

Who bears (consumers or producers) a tax?

Situation: Government made taxes on bottled water

(1)Producer pays for it : less profit, lays off employees to stay alive

(2)Buyer pays for it : increased price, less demand, lays off employees to stay alive

Question is on elasticity of demand and supply

Suppose buyers pays for it.

If buyers still buy as many bottles as before, then demand is inelastic, tax burden goes to buyers.

If buyers buy much less than before, demand is elastic, producers burden the tax.

chapter 4

KEY TERMS

Price controls
Price ceiling
Price floor
Inefficient
Inefficient allocation to consumers
Wasted resources
Inefficiently low quality
Black markets
Minimum wage
Inefficient allocation of sales among sellers
Inefficiently high quality
Quantity control
Quota
Quota limit
License
Demand price
Supply price
Wedge
Quota rent
Excise tax
Incidence
Excess burden
Deadweight loss